BTS–VI(R/S)–05–24–2446 Reg. No. [ ]

A

B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

CE 19-201-0603 ADVANCED METHODS OF STRUCTURAL ANALYSIS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Interpret the concepts of structure-based flexibility matrix method initiated from the compatibility equations in the method of consistent deformation for statically indeterminate plane structures.

CO2: Apply the element-based flexibility matrix approach to analyze rigid-jointed and pin-jointed plane structures.

CO3: Formulate stiffness matrices of basic beam and truss elements and analyze rigid and pin- jointed structures via structure-based and element-based stiffness methods, initiated from the equilibrium equations of the slope-deflection method.

CO4: Appreciate the direct stiffness method as a generalized approach which would in turn seed the concept of the finite element analysis of structures.

CO5: Relate analyze multi-storied rigid-jointed frames by approximate methods for lateral and vertical loads so as to check the output given by any structural analysis software.

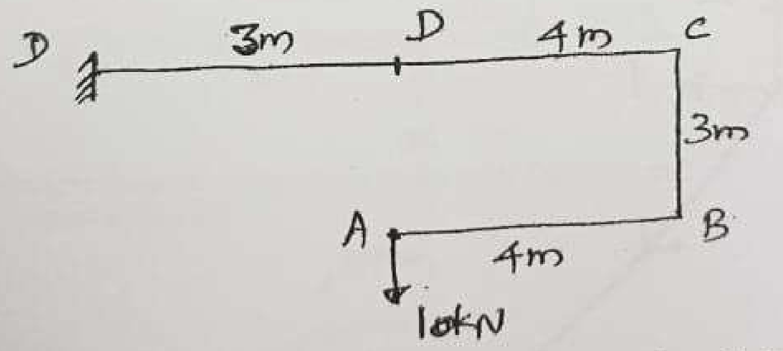
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

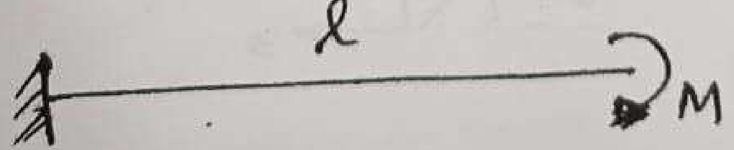
PART A

(Answer *ALL* questions)

		Marks	BL	CO	PO
	(8 × 3 = 24)				
I. (a)	Draw BMD for the frame shown in the figure.	3	L5	1	1

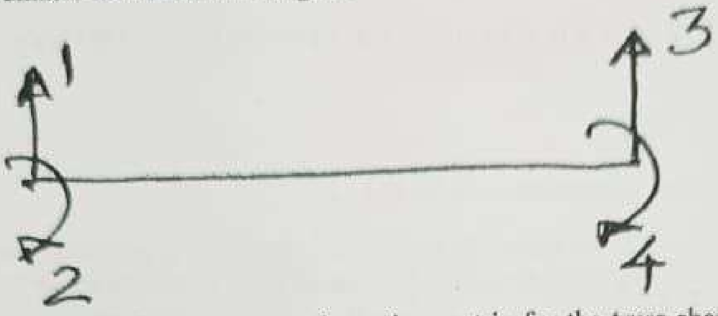


		Marks	BL	CO	PO
(b)	Compute the slope at the free end of the cantilever beam shown in the figure using flexibility method.	3	L6	2	2

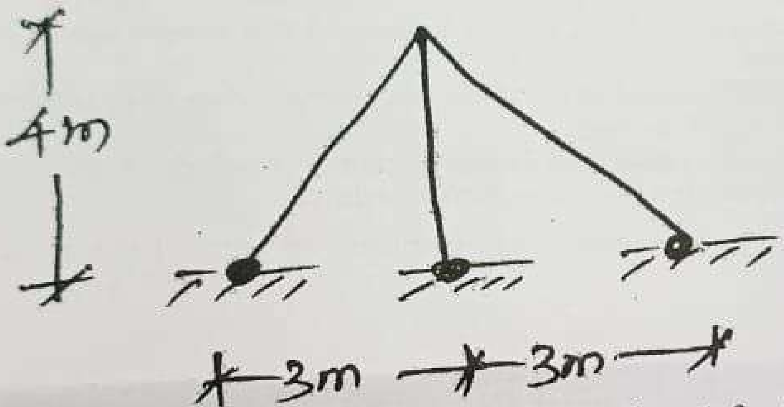


(P.T.O.)

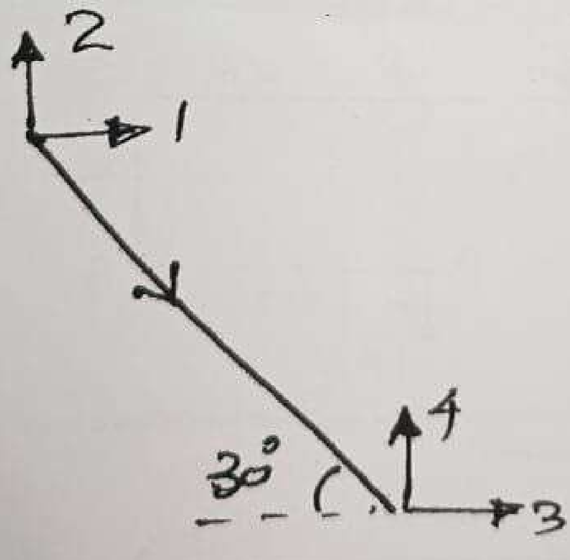
		Marks	BL	CO	PO
(c)	Write down the stiffness matrix for the beam element for the coordinates indicated in the figure.	3	L1	3	2



(d)	Obtain the displacement transformation matrix for the truss shown in the figure.	3	L3	3	2

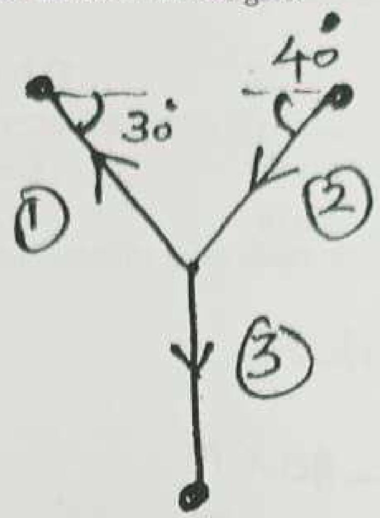


(e)	Obtain the global stiffness matrix for the truss element for the coordinates indicated in the figure.	3	L1	4	1



(Continued)

		Marks	BL	CO	PO
(f)	Draw element or local coordinates for each of the members of the truss, for the incidence indicated in the figure.	3	L2	4	2

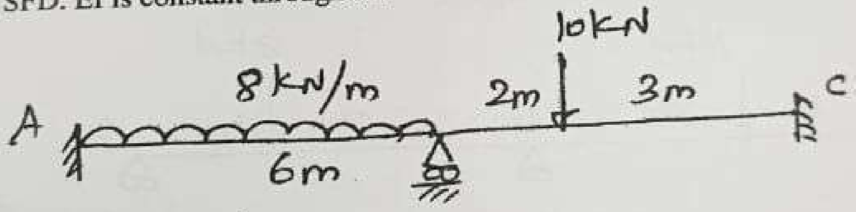


		Marks	BL	CO	PO
(g)	Write down the various assumptions involved in the Portal method.	3	L3	5	2
(h)	Draw substitute frames for finding the negative BM at the support sections of beams, which are located at the extreme top most level of a G+3 storyed building frame, having 3 bays.	3	L1	5	1

PART B

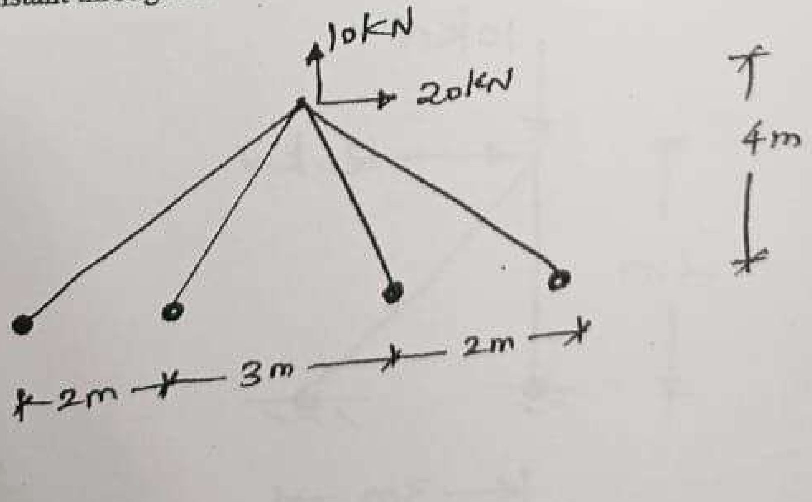
(4 × 12 = 48)

		Marks	BL	CO	PO
II.	Compute the support bending moments at A, B and C using *flexibility* method and draw BMD for the beam shown in the figure. Also draw SFD. EI is constant throughout.	12	L4	2	2

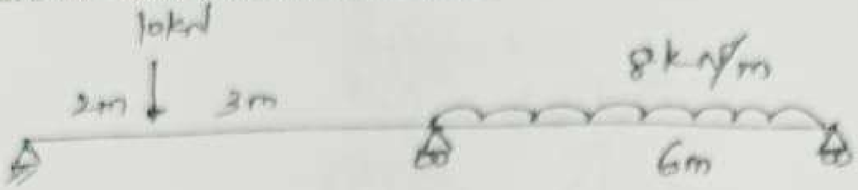


OR

		Marks	BL	CO	PO
III.	Analyse the truss shown in the figure using flexibility method. AE is constant throughout.	12	L4	2	2

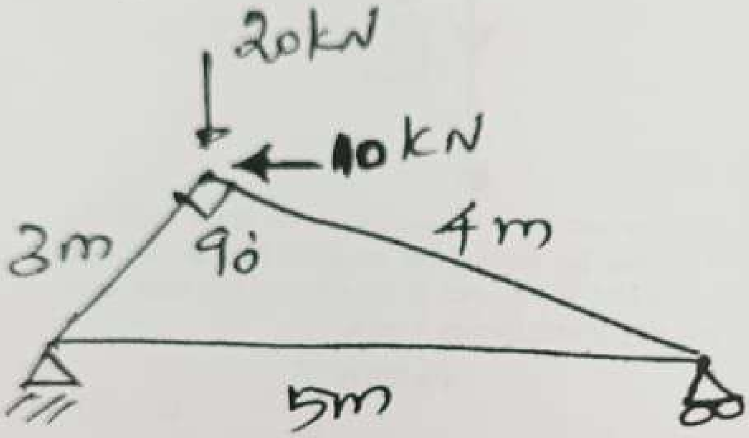


		Marks	BL	CO	PO
IV.	Analyse the beam shown in the figure using stiffness method and draw SFD and BMD. EI is constant throughout.	12	L4	3	2

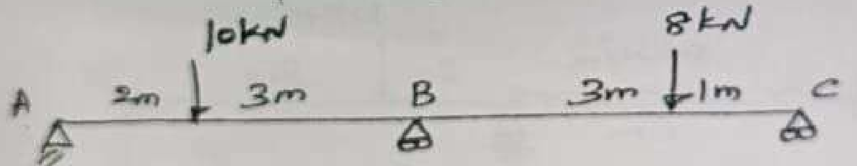


OR

		Marks	BL	CO	PO
V.	Analyse the truss shown in the figure using stiffness method. AE is constant throughout.	12	L4	3	2

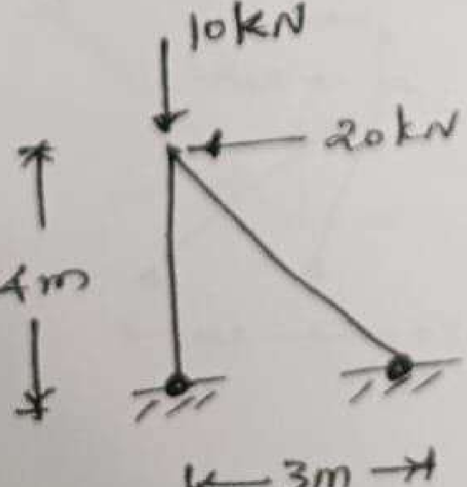


		Marks	BL	CO	PO
VI.	Analyse the beam shown in the figure using direct stiffness method and draw BMD. EI is constant throughout.	12	L4	4	2

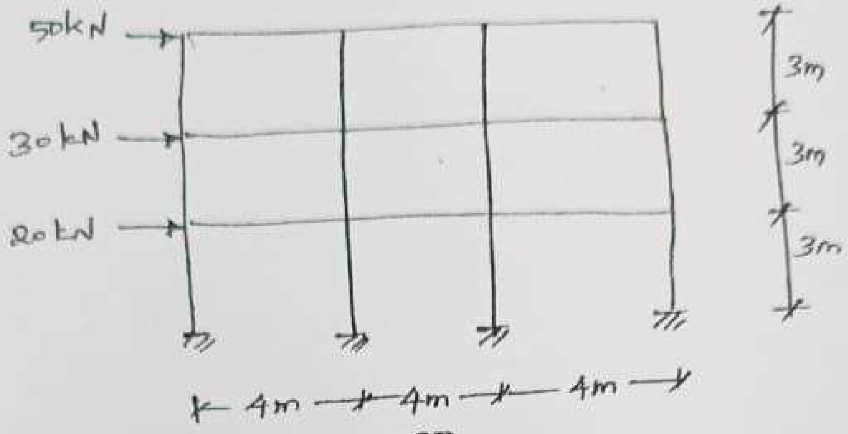


OR

		Marks	BL	CO	PO
VII.	Analyse the truss shown in the figure using direct stiffness method. AE is constant throughout.	12	L4	4	2

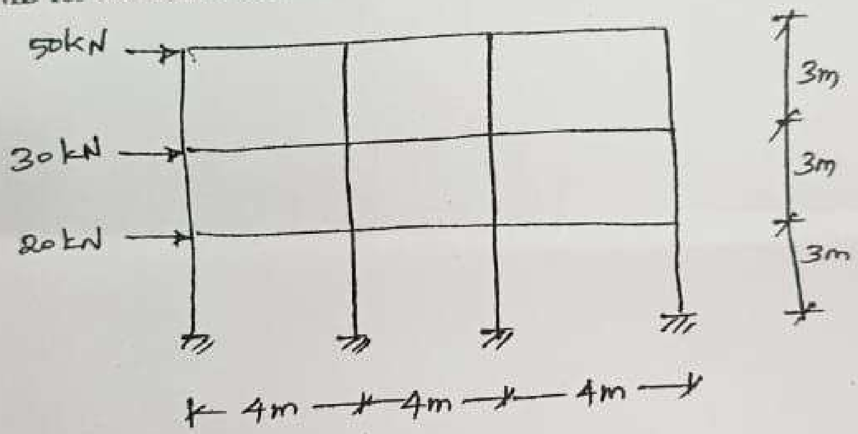


		Marks	BL	CO	PO
VIII.	Analyse the frame shown in the figure using Portal method. Draw BMD for beams and columns.	12	L4	5	2



OR

		Marks	BL	CO	PO
IX.	Analyse the frame shown in the figure using Cantilever method. Draw BMD for beams and columns.	12	L4	5	2



Blooms's Taxonomy Level's
L1 – 7.5%, L2 – 2.5%, L3 – 5.0%, L4 – 80%, L5 – 2.5%, L6 – 2.5%.

***